

LEOPARD GECKO · EUBLEPHARIS
MACULARIUS

Twenty years of gecko, planned from day one.

A complete, fact-checked printable manual: belly heat, the three-hide system, feeder charts, health triage, and the routines that keep a leopard gecko thriving for its full lifespan.

INSIDE

- ✓ Belly heat, humidity and UVB targets in °F and °C
- ✓ Feeder, gut-load and never-feed charts, 60+ rows
- ✓ Stick tail, MBD and impaction triage, plus a symptom table
- ✓ Owner logs, pet-sitter sheet and a power-outage plan

34 pages

Beginner

Print or tablet

Version 2.1 · Sep 2026



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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

If you haven't set up yet

Read pages 4 through 12 in order before buying anything. The enclosure size on page 5 and the three-hide layout on page 6 decide most of what ends up in the cart, and they are the two things that are expensive to get wrong twice. Then use the Budget & Shopping List on page 24 to buy once.

If your leopard gecko is already set up

Jump to the Setup Checklist and Targets on page 22 to audit what you have, and the Health Red Flags section starting on page 16 if anything seems off. Check the warm hide with an infrared thermometer before you change anything else; a warm side that reads too low explains more refused meals than any other single cause.

Print these first

- | | |
|--|---|
| ☐ Page 22: Setup Checklist & Targets | ☐ Page 29: Pet-Sitter Sheet |
| ☐ Page 23: Emergency & Quick Targets Card | ☐ Page 30: Owner Log |
| ☐ Page 26: Symptom Quick Reference | ☐ Page 31: Equipment, Supplement & Vet Log |
| ☐ Page 27: Daily, Weekly & Seasonal Routine | ☐ Page 32: Enrichment Checklist & Log |

WHY THIS PACKAGE EXISTS

Leopard geckos are sold as the easy reptile, and the care sheet that comes taped to the box is usually a decade out of date: a 10 or 20 gallon tank, calcium sand, a single hide, a heat lamp instead of belly heat, and no mention of UVB at all. Every one of those is now either outdated or actively harmful. Cross-checked against veterinary and herpetological sources (listed on page 34), not recycled forum advice. Every temperature and dimension is given in both imperial and metric.

NOT A SUBSTITUTE FOR

A reptile-experienced veterinarian. This is a husbandry reference for day-to-day care. Anything urgent or health-related is always a professional call, not a page in this book. Find one before you need one and write the number on page 23.

SECTION 01

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Eublepharis macularius</i>
Adult size	6.5 to 11 in (17 to 28 cm), sources differ. Males 60 to 90 g, females from about 45 g
Captive lifespan	15 to 20 years typical, up to 25 to 30 recorded. Wild geckos manage 6 to 8
Native range	Pakistan, Afghanistan, Iran, India, Nepal. Rocky desert and dry scrub
Diet type	Obligate insectivore. Live insects only, no fruit or vegetables
Enclosure minimum (adult)	36×18×18 in (91×46×46 cm), floor space over height
Heat source	Under-tank heat mat on a thermostat, 88 to 92°F (31 to 33°C) at the hide floor
UVB requirement	Optional but genuinely beneficial. Low output, UVI 0.5 to 1.5 at the warm hide
Sexual maturity	9 to 18 months
Handling difficulty	Easy once settled, but never by the tail. See page 8
Conservation status	Least Concern (IUCN)

Cost overview

Initial setup

\$175 to \$495 for equipment, plus \$150 to \$215 for the first vet exam and fecal test. The gecko itself is \$20 to \$45 for a wild type, more for a morph.

Monthly ongoing

\$32 to \$91 for insects, supplements, substrate, electricity and averaged vet care. Most owners land between \$250 and \$700 a year all in.

BUDGET FOR THE WORST CASE TOO

An annual wellness exam runs \$85 to \$105. A sick or emergency visit runs \$100 to \$800 or more, and diagnostics or impaction surgery push it higher again. Treat a small emergency fund as part of the setup cost, not an optional extra.

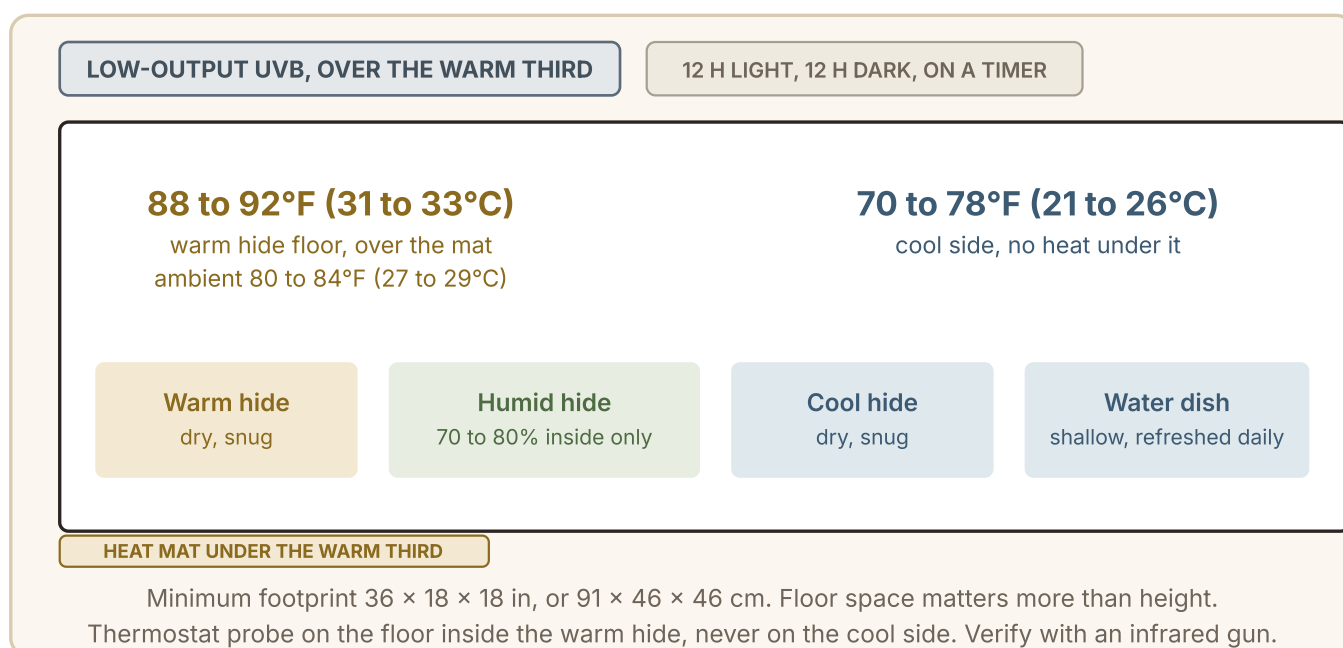
Fun facts worth knowing

- Unlike most geckos, leopard geckos have true eyelids and can blink. They also lack the toe pads other geckos climb glass with.
- That fat tail is roughly a quarter of the gecko's body mass and a third of its length. It is a working fat reserve, which is why tail thickness is the best health check you can do at a glance.
- Over 100 color morphs exist in captivity, and pigment matters practically: pale morphs burn under UVB a normally pigmented gecko handles fine.

Housing & enclosure

36×18×18 in (91×46×46 cm) is the current minimum for one adult, and bigger is always fine. The 20 gallon long tank most older care sheets still name, roughly 30×12×12 in (76×30×30 cm), is no longer considered adequate: it is too short to hold a real temperature gradient from end to end, which is the whole point of the enclosure. Hatchlings can start in something as small as a 10 gallon while they are small, then move up.

Leopard geckos are terrestrial and cannot climb smooth glass, so floor space matters far more than height. Go for a front-opening, well-ventilated enclosure with an escape-proof top. House one gecko per enclosure: they are not a social species, two males will fight, and even females that seem to tolerate each other compete for the warm hide.



NEVER

Run a heat mat without a thermostat. An unregulated mat is a genuine burn and fire risk, and thermal burns from an unregulated heat source are one of the most common injuries a reptile vet sees in this species. Skip hot rocks entirely: they heat unevenly and burn geckos that are sitting on them for warmth.

What the diagram doesn't show

- Hides should be snug enough that the walls touch the gecko's sides. An oversized hide gives no security, and a gecko without security stops using the warm end properly.
- Add rock, slate or cork for nail filing and low climbing. Leopard geckos are not arboreal, but they explore every inch of floor and will use a ledge.
- The humid hide belongs on the cool or middle side. Directly over the mat it dries out in hours, and cooks in between.

Temperature, humidity & the three hides

ZONE	TARGET (°F)	TARGET (°C)
Warm hide floor (belly heat)	88 to 92	31 to 33
Ambient warm side	80 to 84	27 to 29
Cool side	70 to 78	21 to 26
Night drop	65 to 70	18 to 21
Humidity	30 to 40% through the enclosure, 70 to 80% inside the humid hide only	

Measure the warm hide floor with an infrared thermometer, pointed at the surface the gecko lies on rather than the air above it. Use digital probes for ambient air and humidity, mid-enclosure. Stick-on analog dials are inaccurate enough to be worse than nothing.

The three hides are the husbandry, not the decoration

A warm dry hide, a cool dry hide and a humid hide are the minimum, and the humid hide is the one people skip. Line it with damp sphagnum moss or damp paper towel, genuinely damp rather than swampy or bone dry. A gecko uses it in the days before a shed to soften the old skin, and it prevents most of the stuck shed on toes, tail tips and eye caps that becomes a vet visit later. Leopard geckos thermoregulate by moving between zones rather than basking, so an enclosure that is warm end to end is as much of a problem as a cold one.

Humidity: the detail people get wrong

Keep the enclosure itself at 30 to 40% and put the moisture in one box rather than in the air. Raising whole-enclosure humidity to reach 70% is not a substitute for a humid hide, it just adds a respiratory infection risk. If the room is very dry, add a second damp hide rather than misting the tank.

UVB: optional, and worth doing anyway

Leopard geckos do not require UVB the way a bearded dragon does, and were kept without it for decades. The research has moved: controlled work found even brief low-level exposure, as little as two hours a day, measurably raises blood vitamin D3 through the skin. Keep the output low, a UVI of 0.5 to 1.5 at the warm hide and 0.5 to 0.7 for pale morphs. Distance is on page 7.

Signs your setup needs adjusting

YOU NOTICE	LIKELY MEANS
Food refused, but the gecko looks well and the tail is full	Warm side too cool to digest. Check the hide floor with an infrared gun before anything else
Shed stuck on toes, tail tip or eye caps	Humid hide missing, dried out, or on the warm end where it bakes dry
Gecko sits on the cool side and will not use the warm hide	Warm hide too hot, too open, or too big to feel safe in
Condensation on the glass, or a damp smell	Whole-enclosure humidity too high. Fix ventilation before it becomes a respiratory problem

Thermostats, probes & UVB distance

The thermostat is the safety device

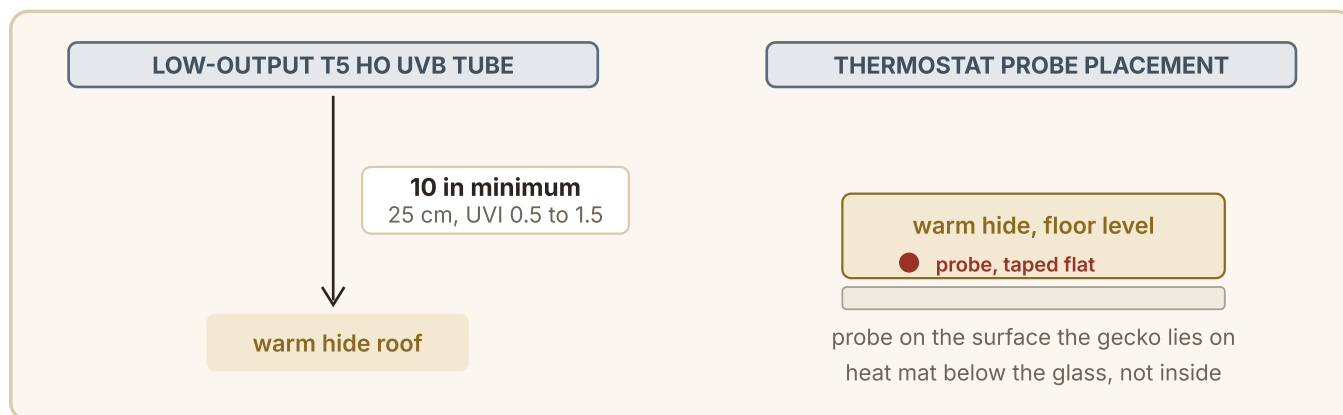
A heat mat has no idea how hot it is. Left unregulated it will climb well past the target on a warm afternoon, and because a gecko seeks belly heat it will stay sitting on the hot spot rather than moving off it. That is how thermal burns happen. An **on/off thermostat** is right for a mat, which changes temperature slowly; a ceramic heat emitter reacts faster and wants a **pulse proportional or dimming** unit that eases output down before it overshoots. Set it to the warm hide floor target, not to the room, and re-check with an infrared gun after any change of season, substrate depth or hide.

Probe placement decides whether the number means anything. The probe belongs flat on the floor *inside* the warm hide, taped down so the gecko cannot shift it, on the same surface the animal lies on. A probe stuck to the glass wall, hanging in the air, or sitting on the cool side reads a different temperature than the one that matters, and the thermostat will happily hold the wrong number all year.

Timers and the light cycle

Put the lights on a plug-in timer so day length never depends on you remembering. Twelve hours on and twelve off is the standard cycle; some care references run the lights up to 14 hours in summer. Skip colored night bulbs entirely, red and blue included: they disrupt sleep in a crepuscular animal. If the room falls below the night floor, use a ceramic heat emitter, which gives heat with no light, on its own thermostat.

UVB distance and probe placement



Distances are for a low-output tube, a 6% or 5% T5 HO, mounted over the warm third. Keep the gecko no closer than 10 in (25 cm) to the lamp, and treat a mesh lid as a 30 to 50% loss of output, which means the tube sits closer to hold the same reading. Higher-output desert bulbs are the wrong tool for this species. Replace UVB tubes every 12 months even if they still light.

MEASURE, DON'T ASSUME

An infrared thermometer costs a fraction of one vet visit and settles the two questions that matter most: what the warm hide floor actually reads, and whether it holds at 3 p.m. on a hot day. Check it monthly and after any change. Keep a second thermometer away from the thermostat probe and read it daily too: a thermostat stuck on, or a probe the gecko has shifted, gives no other warning.

Substrate, furnishings & handling

Substrate

Paper towel or tile are the safest choices and the right default for anyone newer to the hobby: they remove impaction risk entirely and make a stool check trivial. Bioactive soil and sand mixes work, but they are an advanced setup, not a starting point.

NEVER

Calcium sand, or any fine loose sand. It is marketed as digestible and is not; a gecko licking it for calcium or catching a mouthful with an insect is the classic route to impaction. Crushed walnut shell and corncob bedding carry the same risk and add nothing.

Furnishings checklist

- | | |
|---|--|
| ☐ Warm dry hide, snug, directly over the mat | ☐ Shallow water dish the gecko can climb out of |
| ☐ Cool dry hide at the far end | ☐ Slate, rock or cork flat for nail filing and low climbing |
| ☐ Humid hide with damp moss, cool or middle side | ☐ Shallow dish of plain calcium, left in permanently |

Handling

Wait one to two weeks after a new gecko comes home, as a benchmark rather than a countdown: do not start until it has stopped bolting for cover and is eating normally. A gecko under about 3 in (8 cm) waits until it reaches roughly 6 in (15 cm). Then start with 5 minute evening sessions every other day, building to 10 to 20 minutes. Let it walk onto your hand rather than grabbing from above, and stay low so a jump is a short fall.

- **Never lift a leopard gecko by the tail.** They drop it, called caudal autotomy, and the tail is roughly a quarter of the body mass and a third of the length. It regrows in 4 to 8 weeks but comes back shorter, thicker and visibly different.
- No handling during a shed, and none for about 48 hours after a large meal. Chirping, squeaking, rapid tail waving or a fast retreat all mean the session is over.

SALMONELLA

Healthy reptiles shed *Salmonella* without being ill, and one negative test clears nothing. Not a turtle problem: a CDC investigation into pet-gecko infections, closed January 2026, counted 113 cases across 36 states. Wash with soap after every contact with the gecko or its enclosure. Never clean gear in a kitchen sink, a bathroom sink or a bath people use; keep a plastic tub for it. Do not kiss the gecko. Under-fives, over-65s, pregnancy and a weakened immune system are the risk groups, and the CDC line for under-fives is not to handle reptiles at all.

First days in a new home

Run the enclosure for a day or two before the gecko arrives, so the thermostat has been checked at its warmest and coldest. For the first week leave the animal alone apart from fresh water and a daily look: no handling, no rearranging, no worrying about refused meals. Offer food each evening anyway. Hiding all day and refusing food are normal after a move; weight loss over two weeks is not.

Diet & feeding by age

Leopard geckos are obligate insectivores: live invertebrates and nothing else. Their nutrition is decided less by which insect you pick than by what that insect ate first (page 11) and what you dusted it with.

AGE	PREY SIZE	BEST FEEDERS	FEEDINGS	HOW MUCH
Hatchling, 0 to 3 months	About 1/4 in (6 mm)	Black soldier fly larvae, small crickets, small dubia	Daily	What it clears in about 15 minutes
Juvenile, 3 to 10 months	No wider than the space between the eyes	Crickets, dubia roaches, black soldier fly larvae	Daily, or every 1 to 2 days	What it clears in about 15 minutes
Adult, tail as thick as the neck	No wider than the space between the eyes	Dubia roaches, crickets, black soldier fly larvae	Every 2 to 4 days	A few insects, adjusted to body condition

Sources disagree on the adult interval, from every other day to every five. Feed the gecko in front of you rather than the calendar: a tail as thick as the neck and a stable weekly weight mean it is right. Feed in the evening, and use tongs.

Supplement schedule

SUPPLEMENT	JUVENILE	ADULT	NOTES
Plain calcium (no D3, phosphorus-free)	Most feedings	2 to 3 times a week	The daily workhorse. Also keep a shallow dish of it in the enclosure permanently so the gecko can top up on its own.
Calcium with D3	Twice a week	Once or twice a week	Only if the setup has no meaningful UVB. A gecko under a working tube is already making its own D3.
Multivitamin	Once a week	Once a week	Covers vitamin A, which matters for shedding and eye health. Use in place of that feeding's calcium, not on top of it.

NEVER DUST D3 AT EVERY FEEDING

Vitamin D3 is fat-soluble and accumulates, so more is not safer. Plain calcium is the routine dusting; calcium with D3 is a twice-weekly backup for a setup without UVB, and it is one or the other, not both stacked. If you are running UVB and still dusting D3 at every meal, drop the D3 to once a week or stop it.

Water and the calcium dish

Fresh dechlorinated water in a shallow dish the gecko can climb out of, refreshed daily and scrubbed weekly. The separate plain calcium dish stays in permanently: geckos self-regulate from it, and one suddenly eating a lot of it is worth noticing.

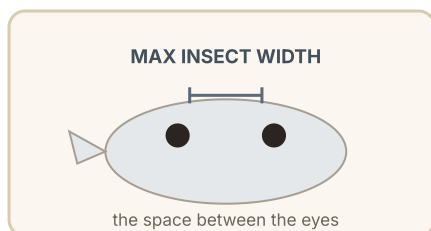
NEVER FEED

Fireflies, which are genuinely toxic, wild-caught insects of any kind, wild hornworms (bright green, as opposed to the safe blue-green captive-bred ones), or any fruit or vegetable. The full never-feed list is on page 12.

Feeder insects: staples, treats & never

Commercially raised only, gut-loaded first, and no wider than the space between the eyes.

INSECT	TIER	NOTES
Dubia roaches	STAPLE	The best all-round feeder here: good protein, lower fat than mealworms, easy to gut-load, quiet, no smell, and they cannot climb smooth plastic.
Crickets	STAPLE	The usual staple, cheap and active, which triggers a real hunting response. Remove any not eaten at lights-out; loose crickets nibble a sleeping gecko.
Black soldier fly larvae (calciworms, Phoenix worms)	STAPLE	The one common feeder that is not calcium-poor: BSFL carry calcium in the exoskeleton and land close to or above a 1:1 calcium to phosphorus ratio unaided. Keep dusting anyway.
Discoid roaches	STAPLE	Same role as dubia, and the usual substitute where dubia are restricted.
Hornworms (captive-bred only)	OCCASIONAL	Mostly water. Genuinely useful for hydration or for tempting a gecko back onto food, poor as a staple. They grow fast, so re-check the size every single day.
Mealworms	OCCASIONAL	Fatty, with a hard chitin shell some geckos struggle to digest in quantity. Fine in rotation, wrong as the main feeder, and the classic route to an obese adult.
Superworms	OCCASIONAL	Adults only. Large, fatty and they bite, so tong-feed and never leave one loose in the enclosure. Not appropriate for juveniles at any size.
Waxworms	TREAT ONLY	Very high fat and habit-forming: a gecko that has had them will hold out for more. A couple a week at most, and none at all for an overweight gecko.
Fireflies and lightning bugs	NEVER	Genuinely toxic to leopard geckos. This is not a moderation issue.
Wild hornworms (bright green)	NEVER	Toxic. The safe ones are captive-bred and appear blue-green. Buy from a reptile supplier, never off a tomato plant.
Any wild-caught insect, spider or centipede	NEVER	Pesticide and parasite risk, however clean the garden looks, and some are venomous or toxic outright.
Fruit, vegetables, or prepared fruit-based gecko powder	NEVER	A leopard gecko is not a crested gecko. It is an obligate insectivore and its gut is not built for produce.



Sizing and presentation

Prey wider than the gap between the eyes is the second most common impaction cause after loose substrate. Hatchlings take insects around 1/4 in (6 mm); after that, measure against the gecko rather than the calendar. Tong-feeding controls the portion, keeps insects out of the substrate, and stops a gecko learning that fingers arrive with food.

Gut-loading: what to feed the feeders

A gecko's nutrition is mostly decided 24 to 48 hours before the insect is ever offered.

Feeder insects are a poor calcium source, and the problem is the ratio rather than the raw amount: phosphorus competes with calcium for absorption, and the accepted target for insectivorous reptiles is roughly 2 parts calcium to 1 part phosphorus, or 1:1 at minimum. Unsupplemented crickets sit near 1:9. Mealworms are worse, in the region of 1:14. An insect is a container, so what is in its gut when the gecko eats it gets eaten too. That is the whole mechanism, and it only works if the diet you put in front of the insect is genuinely calcium-rich.

GUT-LOAD FOOD	TIER	NOTES
Commercial high-calcium gut-load diet	STAPLE	Buy one sold on its calcium content rather than one that just says "gut-load" on the front. The reliable option, and it works for crickets as readily as roaches.
Collard greens	STAPLE	Calcium-dense and low in oxalates, which is the combination that actually moves the ratio.
Mustard greens and turnip greens	STAPLE	Same profile as collards. Rotate them so the insects keep eating.
Dandelion greens	STAPLE	Excellent, and free if the lawn is genuinely unsprayed. If in doubt, buy them.
Butternut squash	STAPLE	Calcium-dense, holds moisture, and insects take it readily.
Calcium-fortified water gel or cubes	STAPLE	Insects drown in an open dish. A fortified gel gives water and calcium through the same route, and insects drink even when they will not eat a dry diet.
Carrot, potato, apple	OCCASIONAL	Keeps insects alive and hydrated, which is worth doing. As a calcium intervention it does close to nothing, so it is not gut-loading on its own.
Spinach, beet greens, Swiss chard	NEVER	They read as healthy and do contain calcium, but they are high in oxalic acid, which binds that calcium into a form nothing can absorb. Self-defeating as a gut-load.

Timing: 24 to 72 hours on the diet, with 48 hours the figure most often cited. Too short and the insect has not eaten enough for its gut contents to matter. Too long stops helping, because gut-loading is not cumulative, an insect only carries what is in its digestive tract right now. Insects start voiding those contents as soon as they leave the food, so feed them out within a few hours of taking them off it. Running a standing colony is simpler: leave the gut-load available all the time rather than treating it as a two-day event before each feeding.

GUT-LOAD AND DUST, NOT ONE OR THE OTHER

A properly formulated high-calcium gut-load can in principle get an insect to an acceptable calcium to phosphorus ratio on its own. In practice, dust as well: you cannot see how much of the diet a given cricket actually ate, and dusting is the cheap insurance that covers the gap. Page 9 has the schedule.

Treats, refusals & the never-feed list

What to offer beyond the staples, and how to read a gecko that has stopped eating.

ITEM	VERDICT	NOTES
Waxworms, offered by tong	TREAT ONLY	A couple a week at most. Genuinely useful for getting a gecko to take an oral medication.
Hornworms, captive-bred	OCCASIONAL	The best tool for a mildly dehydrated or off-food gecko, because they are mostly water.
Pinky mice	RARE AT MOST	Listed as an occasional item in some veterinary care sheets. Not needed by a well-fed insectivore, and easy to overdo.
Fruit, vegetables, baby food, honey	NEVER	Obligate insectivore. The digestive system is not built for produce, whatever a video suggests.
Crested gecko diet or other fruit-based powders	NEVER	A complete diet for a different species. It does not substitute for insects here.

Why a leopard gecko stops eating

1. **The warm side is too cool.** The most common fixable cause by a distance. Check the warm hide floor before anything else.
2. **A shed is coming.** Many geckos go off food for a day or two either side of one. Appetite returns when it is done.
3. **Brumation.** A natural winter slowdown cued by cooler temperatures and shorter days, with reduced activity and brief waking to drink.
4. **Stress.** A new enclosure, too much handling, or an incompatible tankmate. It resolves when things settle.
5. **Breeding season.** Both sexes may eat less during courtship, and gravid females often eat less as they near laying.
6. **Prey boredom.** Some geckos lose interest in a repeated feeder and pick straight back up on something different.
7. **Illness or impaction.** Refusal alongside bloating, straining, loose stool, poor shedding or sunken eyes points at a medical cause, not a cycle.

WHEN A REFUSAL BECOMES A VET VISIT

Watch the tail, not the calendar. Estimates for "still probably fine" in a healthy adult run from about two weeks to a month, and no one agrees on a day count. A tail that has visibly thinned over a couple of weeks is the concrete signal. Go sooner for lethargy beyond normal brumation behavior, bloating, straining, incomplete shedding, sunken eyes, or wrinkled skin. Juveniles have far less margin.

THE NEVER-FEED LIST, IN ONE PLACE

Fireflies and lightning bugs. Wild-caught insects, spiders and centipedes of any kind. Wild hornworms, the bright green ones. Fruit, vegetables, purees and baby food. Crested gecko diet and other fruit-based powders. Anything larger than the space between the gecko's eyes. Anything from a bait shop.

Common mistakes & enrichment

Mistakes that cause the most harm

- **One hide instead of three.** A single hide forces the gecko to choose between feeling safe and being the right temperature, and it removes the humid hide that prevents most shedding problems.
- **A heat mat with no thermostat, or a probe in the wrong place.** Both produce the same result: a surface that reads fine when you check it and climbs past 100°F (38°C) when you do not.
- **Mealworms and waxworms as the staple diet.** The fastest route to an obese adult, and obesity in this species shows up as fat pads at the neck and armpits long before the owner calls it a problem.

Enrichment, by priority

Leopard geckos are one of the few reptiles with real published enrichment research behind them, and it is specific enough to change what is worth buying. Enrichment comes after correct belly heat, hides and humidity, never instead of them. In this order:

1. Thermal choice and the three hides

Thermal enrichment produced one of the two strongest responses in the Bashaw study. A warm dry hide, a cool dry hide and a maintained moist hide are what let a gecko pick its own microclimate without giving up cover. This is the foundation; nothing below compensates for missing it.

2. Feeding and hunting presentation

The other strong result. Tong-feeding, short supervised insect hunts inside the enclosure, a rolling dispenser that sheds insects as it is nudged, and simple timed-release prey containers all beat a static bowl. Feed in the evening, when the animal is naturally hunting.

3. Low climbing structure and cover

Cork flats and tubes, low sturdy branches and stacked slate give somewhere to rest off the floor and visual cover that still allows movement. Height matters far less here than for an arboreal species.

4. A dig box

A single box of damp, safe substrate satisfies real digging behavior, which shows up especially in some females, without turning the whole floor into loose substrate. Paper towel or tile stays the safe primary surface.

5. Light novelty rotation

A new piece of cork, a mild rearrangement. Some value, genuinely last on the list, and not a substitute for anything above it. Rotate monthly rather than weekly; a gecko that has to re-learn its enclosure every few days is not being enriched.

6. What the research says does not work

Mirrors and other visual enrichment were the weakest of the five types tested. Not harmful, just close to pointless, and the same goes for exercise wheels. Geckos in enriched housing showed better welfare indicators than those in sparse setups, and chose the enriched side in preference tests, so the answer is more of items 1 to 4, not novelty for its own sake.

Sexing, growth & body condition

Telling male from female

Two features, both just in front of and behind the vent, and both easier to read as the gecko approaches maturity at 9 to 18 months. **Males** show a row of enlarged preanal pores in a V shape ahead of the vent, which produce a waxy secretion used for scent marking, and a pair of hemipenal bulges at the tail base just behind it. **Females** have pores that are much smaller or absent and no bulges. Lift the tail gently to about 45 degrees in good light rather than restraining the animal, and never hold the tail itself. Hatchling sexing at a shop is a guess; recheck at six months and again at a year.

Growth reference

Ranges are deliberately wide and healthy geckos fall outside them. What matters is the trend: steady gain for a juvenile, a stable weight for an adult. Weigh weekly in grams on a kitchen scale and log it on page 30.

AGE	TOTAL LENGTH	WEIGHT
Hatchling	3 to 4 in (8 to 10 cm)	2 to 8 g, sources differ
2 months	About 5 in (13 cm)	18 to 30 g
6 months	5 to 6 in (13 to 15 cm)	25 to 60 g
18 months and up (adult)	8 to 11 in (20 to 28 cm)	40 to 80 g
Adult male vs female	Males 8 to 10 in, females 7 to 8 in	Males 60 to 90 g, females from about 45 g

Published adult lengths run from 6.5 to 11 in (17 to 28 cm) depending on the source, which is a real disagreement rather than a typo; see page 34. Most geckos reach full length by about 18 months and keep filling out for another six months after that.

Body condition at a glance

UNDERWEIGHT

Tail narrower than the neck, or visibly thinning week to week. Hip bones and spine standing out, sunken eyes, loose skin when there is no shed due. Rule out low warm-side temperature, parasites and cryptosporidiosis (page 17) before simply feeding more.

IDEAL

Tail as thick as the neck at its base, full and rounded, with no bony ridge along the spine or hips. Skin smooth, eyes full, weight stable week to week in an adult and climbing steadily in a juvenile.

OVERWEIGHT

Fat deposits at the neck and in the armpits, a tail noticeably wider than the head, belly wide from above. Almost always a mealworm and waxworm diet. Cut the fatty feeders and stretch the interval, do not fast the gecko.

WHEN A WEIGHT CHANGE IS A VET VISIT

A tail that is thinning while the gecko is still eating is the one that needs an appointment quickly, because that pattern is the signature of cryptosporidiosis (page 17). Any steady loss across three or four weekly weigh-ins is worth a call, as is a juvenile that stops gaining. Bring the Owner Log; a trend line tells a vet more than a single reading.

Females, eggs & egg binding

A female kept entirely alone can still produce eggs. This page is for every female owner, not just breeders.

Eggs without a male

Female egg-laying reptiles can produce eggs with no male present, so every female is at risk of being unable to pass one. Those eggs are infertile and will never hatch, but the physical process is identical, and so is the calcium demand behind it. A mature female can lay several clutches in a season, typically two eggs at a time with roughly a month between clutches, and a female in that cycle draws heavily on her calcium reserves. This is the practical reason the plain calcium dish stays in the enclosure permanently.

Signs a female is gravid or about to lay

- Two pale, oval shapes visible through the skin of the belly, most obvious when she is on glass or held briefly over a light.
- Restlessness and repeated digging, especially in a corner or in the humid hide.
- Appetite dropping off in the days before laying, then returning afterwards.

Give her somewhere to lay

Inadequate nesting is a recognized cause of egg retention in lizards, and it is the easiest one to remove. Put a lay box in as soon as she looks gravid: a plastic tub with an entry hole cut in the lid, filled 3 to 4 in (8 to 10 cm) deep with damp, holdable substrate such as coco fiber or damp sphagnum, kept on the warm or middle side. Damp enough to hold a burrow shape, not wet. Leave her alone with it, since disturbance during laying is exactly what makes a female hold on to eggs.

AFTER SHE LAYS

Remove infertile eggs; they mold. Offer food and water, and expect appetite to bounce back within a day or two. Weigh her: a laying female loses weight, and knowing how much this time makes the next cycle readable. If she is producing clutch after clutch, ask a reptile vet about it rather than treating it as normal, because repeated laying is a genuine calcium drain.

Egg binding (dystocia) is a same-day call

Egg binding is a female straining but unable to pass eggs. Laying should be complete in under 48 hours; past that, or at any point she looks stressed or unwell, it needs a vet rather than more waiting. Causes include an inadequate nest site, poor calcium status, oversized or misshapen eggs, illness, and incorrect temperature or humidity. Treatment ranges from correcting the nest site through calcium and hormone injections to surgical removal, and the earlier it is caught the further down that list it stops.

- Straining, a swollen or protruding cloaca, restlessness that does not settle, repeated digging with nothing produced.
- Lethargy, depression, refusing food while still visibly gravid, or any tissue protruding from the vent, which is an immediate emergency.

Health red flags

What to watch for, and what to actually tell the vet.

CONTACT A REPTILE-EXPERIENCED VET PROMPTLY IF YOU SEE

A tail thinning to a stick, with or without weight loss. A soft, spongy or visibly deformed jaw. Tremors or twitching limbs. Any breathing difficulty, wheezing, clicking, or open-mouth breathing at rest. Drooling, or pus or cheesy material around the mouth. Appetite loss that persists more than a few days, and any refusal at all in a juvenile. Tissue protruding from the vent. Retained shed that is swollen, darkened, or will not clear from the eyes or toes.

What to tell the vet

- ☐ The warm hide floor temperature, measured with an infrared gun, not the thermostat's setting
- ☐ Substrate, and whether loose substrate has ever been used
- ☐ Exact supplement routine: which powders, how often, and whether UVB is running
- ☐ When symptoms started and whether they're worsening
- ☐ Last normal bowel movement, and what it looked like
- ☐ Weight trend from the Owner Log, and a fresh stool sample in a sealed bag if you can

Why temperature comes up first

A leopard gecko cannot generate its own body heat, so the warm side is not a comfort setting, it is what powers digestion and the immune response. A warm hide reading a few degrees low produces refused meals, slow or absent stools, an animal that hides all day, and eventually a real infection, and every one of those looks like a different problem. That is why a vet asks for the number before anything else, and why the first thing to check at home is the surface the gecko lies on. The Association of Reptilian and Amphibian Veterinarians keeps a searchable directory of reptile-experienced vets; find one before you need one and write the number on page 23.

Normal vs. concerning

NORMAL	CONCERNING
Dull, greyish skin and hiding for a day or two, then a shed	Dull skin for over a week with no shed, or shed stuck on toes, tail tip or eyes
Reduced appetite in cooler months in an adult, with a stable weight	Reduced appetite with a tail that is visibly thinning, at any time of year
Sleeping in the warm hide most of the day, active at dusk	Lethargic during the evening activity window, or hard to rouse at all

Stick tail disease & metabolic bone disease

Cryptosporidiosis, or "stick tail disease"

Cause: a parasite, most often *Cryptosporidium varanii*, spread between geckos by contact and by feces. It is severe, highly contagious, and currently has no cure. Leopard geckos are the most commonly diagnosed reptile for it, and it is estimated that around half of captive leopard geckos may carry it, many with no symptoms at all. **Signs:** a tail that thins out to a stick, weight loss and emaciation despite a normal or even increased appetite, undigested food in the stool, foul-smelling feces, and a bloated or distended belly. **Diagnosis:** a fecal PCR test through a reptile-experienced vet. False negatives happen, so one clear test does not fully rule it out. **Response:** isolate the gecko immediately, with separate tools and hands washed between animals, and get the test done. **Recovery:** the main drug, paromomycin, manages the condition rather than curing it. Mortality for symptomatic cases runs around 50%, and a gecko that has not responded within a month generally has a poor outlook.

THE PREVENTION THAT ACTUALLY WORKS

Quarantine every new gecko for **3 to 6 months** before it shares a room, a tool, or a hand with an existing animal. That is the Merck figure for a new reptile; 60 to 90 days is the hobby floor. A separate **room** rather than a second tank on the same shelf, since two tanks in one room share air. Paper towel substrate throughout, which is not only easy to strip: it is the diagnostic surface, the thing that makes mites, an abnormal stool and a regurgitated meal visible where loose substrate hides all three. Its own tongs and dishes, and always fed and cleaned last. Asymptomatic carriers spread this without ever looking sick, which is exactly why quarantine, not observation, is the control.

Metabolic bone disease (MBD)

Cause: chronic calcium or vitamin D3 deficiency, and it is common in this species precisely because geckos are so often kept without UVB, without consistent supplementation, or without adequate heat. All three contribute, and this species appears to rely on UVB and correct temperature to process calcium rather than on dietary supplementation alone. **Signs:** trembling or twitching limbs, difficulty walking, a soft or deformed lower jaw sometimes called rubber jaw, pathological fractures, twisted or bent limbs, difficulty eating, lethargy and weight loss. **Diagnosis:** a vet exam with bloodwork and x-rays. **Recovery:** caught early, correcting calcium, D3 and heat reverses the course; advanced cases leave permanent skeletal deformity even after treatment.

CONCRETE THRESHOLD

Any tremor, any softness or asymmetry in the jaw, or any new difficulty walking is a vet appointment this week, not a wait-and-see. In the meantime handle the gecko as little as possible and take out anything it can climb and fall from: weakened bone fractures on a drop that would normally mean nothing.

Impaction, stuck shed & respiratory infection

The three husbandry-driven problems, in the order they turn up.

Impaction

Usually caused by swallowed loose substrate, most often sand, or by prey too large to pass safely. Low warm-side temperature, dehydration and parasites all make it likelier, because a gecko that cannot digest at speed cannot move a mouthful of substrate along either. Watch for a bloated abdomen, appetite loss, weight loss and no normal bowel movement. Warm shallow baths and very gentle belly massage help mild cases; a true blockage is a surgical problem. That surgery is the reason paper towel and tile are the recommended substrate for anyone who is not already confident, and the reason prey width is measured against the gap between the eyes rather than guessed.

Stuck shed (dysecdysis)

Caused by low humidity or a vitamin A deficiency, and it collects in the same three places every time: the eye caps, the toes, and the tail tip. A ring of retained shed around a toe acts as a tourniquet and can cut off circulation completely, which is how geckos lose digits. Never peel dry shed off. Soak the gecko in shallow lukewarm water for 10 to 15 minutes, then work the softened skin off with a damp cotton swab, and repeat over a couple of days rather than forcing it in one go. Retained eye caps, a gecko rubbing its face or holding an eye closed, are a vet job, not a home one. A maintained humid hide prevents nearly all of this, and a check of the toes, tail tip and eyes after every shed catches the rest early.

Respiratory infection

Far less common in a correctly kept leopard gecko than in a damp one, which is exactly why raising whole-enclosure humidity to fix a shedding problem is the wrong move. The cause is nearly always cold, damp, or both: a cool warm side, poor ventilation, or an enclosure misted like a rainforest. Signs are clicking or wheezing when the gecko breathes, open-mouth breathing at rest, mucus or bubbles around the nose or mouth, and lethargy. This is a vet visit rather than a husbandry fix on its own, but fix the husbandry at the same time: get the warm side to target, drop enclosure humidity back to 30 to 40%, and confine the moisture to the humid hide.

THE COMMON THREAD

All three of these are readings, not bad luck. Impaction traces back to substrate and prey size, stuck shed to one damp box, respiratory infection to temperature and ventilation. Any gecko showing one of them should have its warm hide floor, cool side and enclosure humidity measured the same day, because the fix is usually in those three numbers.

Parasites, eye caps, tail loss & other problems

Less dramatic than the last two pages, and far more common.

Internal parasites

Most geckos carry a low parasite load without trouble; it becomes a problem when the animal is stressed, cold or already ill. Loose, foul or unusually frequent stool, undigested food, weight loss with a normal appetite, and a gecko that fails to thrive are the signs. Diagnosis is a fecal exam, which is why a fresh stool sample in a sealed bag turns a first vet visit into a real baseline. Bring one to the wellness exam even if nothing looks wrong: several of the conditions in this section, cryptosporidiosis especially, are present long before they are visible.

Retained eye caps and eye problems

A gecko holding one eye closed, rubbing its face on the substrate, or with a visibly cloudy eye outside the pre-shed window needs looking at. The usual cause is retained shed under the eyelid, sometimes with a vitamin A deficiency behind it, and it can progress to infection. Do not attempt to remove anything from an eye at home. A weekly multivitamin that includes vitamin A is the preventive half of this.

Cystic calculi (bladder stones)

Listed among this species' common presentations in veterinary references. It shows as straining, a swollen lower abdomen, appetite loss and difficulty passing stool or urates, and it is diagnosed on x-ray. It is not something to manage at home, but it is worth knowing the name, because "straining with nothing produced" in a male gecko is easy to misread as constipation.

Tail loss and regrowth

A dropped tail is a stress response, not an injury you caused by accident, and it happens when a gecko is grabbed by the tail, badly frightened, handled roughly or attacked by a cagemate. Move the gecko onto clean paper towel immediately, keep the enclosure especially clean while the stump heals, skip handling for one to two weeks, and watch the stump for swelling, discharge, discoloration or a foul smell, any of which mean a vet. The tail regrows in 4 to 8 weeks, shorter and thicker than the original, and the gecko is otherwise fine. The real cost is the fat reserve it carried, so keep a closer eye on weight for the next couple of months.

Burns and prolapse

Burns come from unregulated heat, hot rocks, or a mat with a failed thermostat, and they present as blistered, reddened or blackened skin on the belly and feet. Turn the heat source off, fix the regulation, and get it seen: reptile burns become infected easily. A prolapse, pink or red tissue protruding from the vent, is an emergency in every case. Keep the tissue moist with clean lukewarm water or sterile saline, do not push it back in, and go the same day.

Shedding, brumation & behavior

The shed cycle

Shedding, properly ecdysis, is how a reptile grows and clears old or damaged skin. The pre-shed phase runs roughly 5 to 14 days: colors go dull and greyish, the skin looks dry, and the gecko hides more and often goes off food. Then it comes off in a day, usually at night, and most leopard geckos eat the shed as they go, so an owner who never sees one is not missing anything. Humidity is the single biggest factor in whether it comes off cleanly, and rough surfaces, cork, slate, a textured hide, give the gecko something to rub against to start the peel.

SHED WEEK ROUTINE

Check the humid hide is genuinely damp and top it up. Leave the gecko alone: no handling during a shed. Skip a feeding if it is refused rather than pushing food. The day after, check the toes, the tail tip and both eyes in good light. Anything still attached gets a lukewarm soak, not a fingernail.

Brumation

A natural winter slowdown cued by cooler temperatures and shorter daylight, not something you have to induce. A brumating gecko is less active, eats little or nothing for weeks, and wakes briefly to drink. That is normal in a healthy adult, and juveniles should not be allowed to brumate deeply. The one rule that keeps it safe: weigh the gecko every couple of weeks through it. A brumating gecko lives off its tail reserve and should not be losing significant body mass, so real weight loss during a suspected brumation is a vet visit, not a season.

Sounds and body language

Leopard geckos are more vocal than most lizards. A bark or chirp is not automatically distress and often just means the gecko wants food or wants you further away. A squeak, a scream (most common in juveniles), rapid tail flicking or waving, fast retreats and refusing food afterwards all mean a handling session has gone on too long. A slow, deliberate tail wave is different: it is the stalking and territorial display, not a complaint. Learning the difference is most of what makes handling go well.

QUICK DIFFERENTIATOR

Brumation looks like a gecko that is quiet, hiding and off food but holding its weight and still bright-eyed when disturbed. Illness looks like the same behavior with a thinning tail, sunken eyes, or a gecko that is hard to rouse at all. Weight is what separates them, which is the whole reason for the log on page 30.

Reading poop & keeping a gecko hydrated

Stool is the earliest free health check you get. Look at it before you clean it up.

IT LOOKS LIKE	USUALLY MEANS	DO
A firm dark brown or black pellet with a soft white urate attached	Normal	Remove it, note it in the log, carry on
Runny, foul-smelling, or more frequent than usual for two days or more	Parasites, a change of feeder, or stress	Collect a fresh sample in a sealed bag and book a fecal test (page 19)
Recognizable insect parts passed whole	Food is moving through undigested: warm side too cool, or a parasite load	Measure the warm hide floor the same day, then vet if the temperature is correct (page 17)
Nothing at all for 5 days in an adult, or 3 in a juvenile	Impaction, dehydration, or a cold warm side	Warm shallow soak, check temperatures, vet if it continues (page 18)
Urate hard, yellow or orange instead of soft and white	Dehydration, and left uncorrected a urate or kidney problem	Soak, refresh the humid hide, offer a hornworm. Persisting is a vet visit: the same uric acid chemistry causes gout

Hydration

A desert species still drinks. Most of a leopard gecko's water comes from the dish and from the humid hide it sits in before a shed, and the rest from its food, which is why a mostly-mealworm diet is a hydration problem as well as a fat one.

- Shallow dish of fresh dechlorinated water, changed daily and scrubbed weekly. Shallow enough that the gecko can climb out without effort.
- The humid hide is hydration equipment, not just shed equipment. Keep the moss damp year round, not only when a shed is due.
- A captive-bred hornworm or two is the quickest way to get water into a gecko that will not drink, since they are mostly water.

SIGNS OF DEHYDRATION

Skin that stays wrinkled or tents when gently pinched with no shed due, sunken eyes, sticky or stringy saliva, hard yellow urates, and a general drop in activity. A 15 to 20 minute soak in shallow lukewarm water, no deeper than the gecko's elbows and never unattended, plus a corrected humid hide, resolves a mild case. A gecko that stays wrinkled after a soak needs a vet, because dehydration that persists usually has something else behind it.

Setup checklist & targets

Essential setup checklist

- ☐ 36×18×18 in (91×46×46 cm) enclosure, front-opening
- ☐ Under-tank heat mat, under the warm third
- ☐ Thermostat, with the probe inside the warm hide
- ☐ Three hides: warm dry, cool dry, humid
- ☐ Sphagnum moss for the humid hide
- ☐ Infrared thermometer, plus a digital probe thermometer and hygrometer
- ☐ Paper towel or tile substrate
- ☐ Shallow water dish and a permanent plain calcium dish
- ☐ Plain calcium, calcium with D3, and a multivitamin
- ☐ Low-output UVB on a timer (optional, recommended)

Temperature & climate targets

TARGET	°F	°C
Warm hide floor (belly heat)	88 to 92	31 to 33
Ambient warm side	80 to 84	27 to 29
Cool side	70 to 78	21 to 26
Night drop	65 to 70	18 to 21
Humidity	30 to 40% enclosure, 70 to 80% inside the humid hide	
Light cycle	12 hours on, 12 hours off, on a timer. No colored night bulbs	

Before your gecko comes home

- ☐ Enclosure running for 48 hours, warm hide floor verified at 88 to 92°F (31 to 33°C)
- ☐ Humid hide filled and damp, all three hides in place
- ☐ A week of gut-loaded insects and both calcium powders bought in advance
- ☐ Reptile vet found and first appointment booked (page 25)

Emergency & quick targets

CALL A REPTILE VET NOW IF

The tail is thinning to a stick. The jaw is soft or crooked, or the limbs are trembling. Breathing is audible, open-mouthed, or there is mucus at the nose. Tissue is protruding from the vent. A female has been straining for more than 48 hours. The skin is blistered or blackened. The gecko is limp or cannot be roused.

READING	°F	°C
Warm hide floor (belly heat)	88 to 92	31 to 33
Cool side	70 to 78	21 to 26
Act below (outage floor)	60	16
Humidity	30 to 40% enclosure, 70 to 80% in the humid hide	

Your numbers

Reptile vet _____

UVB bulb installed _____

24-hour exotic emergency clinic _____

Thermostat last verified _____

Gecko's name and hatch date _____

Last vet visit _____

Routine, written down

Feeding days _____

Multivitamin day _____

Plain calcium days _____

Lights on and off at _____

HAVE THIS READY WHEN YOU CALL

Current weight in grams and the last three from the log. The warm hide floor reading taken with an infrared gun in the last hour. Substrate type. Supplement routine and whether UVB is running. When the gecko last ate and last passed stool. A photo of anything visible.

Budget & shopping list

US retail ranges. The low end is basic but correct gear; the high end is premium equipment.

INITIAL SETUP	TYPICAL COST
Enclosure, 36×18×18 in (91×46×46 cm) minimum	\$50 to \$250
Under-tank heat mat and thermostat	\$50 to \$70
Low-output UVB fixture and bulb (optional, recommended)	\$20 to \$50
Infrared thermometer, digital thermometer and hygrometer	\$15 to \$40
Substrate: paper towel or tile	\$10 to \$25
Three hides and a shallow water dish	\$30 to \$60
Equipment subtotal	\$175 to \$495
First vet exam and fecal test	\$150 to \$215
All-in before the gecko arrives	\$325 to \$710

The gecko itself is separate: \$20 to \$45 for a wild type from a shop, \$50 to \$150 from a reputable breeder even for a basic morph, \$75 to \$300 for popular morphs, and \$500 to \$3,500 or more for rare designer morphs, plus \$35 to \$100 for live shipping if you buy online. Most first-time setups land around \$250 to \$400 once you stop pricing every line at its extreme.

MONTHLY ONGOING	TYPICAL COST
Feeder insects	\$10 to \$30
Calcium and multivitamin powders	\$2 to \$10
Substrate, cleaning and miscellaneous	\$4 to \$20
Electricity for the heat mat	\$5 to \$15
UVB bulb replacement, averaged over the year	\$4 to \$7
Annual wellness exam, averaged	\$7 to \$9
Monthly ongoing	\$32 to \$91

Most owners land between \$250 and \$700 a year all in. Buying feeders in bulk, or running a small dubia colony, is the single biggest lever on that number.

PLAN FOR, DON'T COUNT ON AVOIDING

A sick or emergency visit runs \$100 to \$800 or more, and diagnostics, x-rays or impaction surgery climb from there. A few hundred dollars set aside matters as much as any item on this list.

WHERE NOT TO CUT CORNERS

The thermostat, the enclosure size, and the first vet exam. A cheap enclosure can be upgraded later; a mat run without a thermostat burns a gecko once and there is no version of that you get back.

First 30 days checklist

Day 1

- ☐ Weigh the gecko in grams and photograph it and the setup for a baseline
- ☐ Verify the warm hide floor with an infrared gun before the gecko goes in, then leave it alone
- ☐ Fill in the numbers on page 23 and the equipment dates on page 31

Week 1

- ☐ No handling at all. Fresh water daily, offer food each evening, remove what is not eaten
- ☐ Book a wellness visit with a reptile vet for within the first two weeks, and take a fresh stool sample
- ☐ Confirm the gecko is using the warm hide. If it never does, re-check the temperature and the hide size

Week 2 to 3

- ☐ Once it stops fleeing when you approach and is eating normally, start 5 minute handling sessions every other day
- ☐ Watch for the first shed and check toes, tail tip and eyes the day after
- ☐ Start the Owner Log on page 30 with a weekly weigh-in

Week 4

- ☐ Review the weight trend. Steady gain is the goal for a juvenile, stable weight for an adult
- ☐ Settle the feeding interval on body condition rather than the calendar, using page 14
- ☐ Introduce the first enrichment items from page 32

Days 31 to 90

- ☐ Book any follow-up the vet recommended, and act on the fecal result rather than filing it
- ☐ If this gecko is joining an existing collection, hold the full 3 to 6 month quarantine before they share a room, tools or hands (page 17)
- ☐ Look back at the Owner Log for any slow drift in weight, appetite, or tail thickness
- ☐ Start rotating enrichment on the monthly schedule from page 32, and fill in the pet-sitter sheet on page 29 before you need it

Symptom quick reference

SYMPTOM	LIKELY CAUSE	ACTION
Tail thinning to a stick, appetite normal or increased	Cryptosporidiosis	Isolate, vet, fecal PCR (page 17)
Tremors, soft or crooked jaw, difficulty walking	Metabolic bone disease	Vet this week, remove anything climbable (page 17)
Bloated belly, no stool, straining	Impaction	Warm soak, check temperatures, vet promptly (page 18)
Shed stuck on toes, tail tip or eye caps	Humid hide missing or dried out	Lukewarm soak and swab, never dry peeling (page 18)
Clicking or wheezing, open-mouth breathing at rest	Respiratory infection	Vet; correct temperature and drop enclosure humidity (page 18)
Runny or foul stool for two days or more	Internal parasites, feeder change, stress	Vet with a fresh sealed sample (page 19)
One eye held closed, face rubbing, cloudy eye	Retained eye cap, vitamin A deficiency, infection	Vet, do not touch the eye (page 19)
Straining with nothing produced, swollen lower belly	Impaction or bladder stone	Vet, x-ray (pages 18 to 19)
Blistered, reddened or blackened belly skin	Thermal burn from an unregulated heat source	Turn off the heat source, fix the thermostat, vet (page 19)
Pink or red tissue from the vent	Prolapse	Keep moist, vet the same day (page 19)
Food refused, gecko otherwise bright, tail full	Cool warm side, shed coming, or brumation	Measure the warm hide floor first (pages 12 and 20)
Female restless, digging, off food, two shapes in the belly	Gravid	Lay box now, vet if straining passes 48 hours (page 15)
Skin wrinkled or tenting, urates hard and yellow	Dehydration	Shallow soak, refresh the humid hide (page 21)
Fat pads at the neck and armpits, tail wider than the head	Obesity, usually a mealworm or waxworm diet	Cut fatty feeders, stretch the interval (page 14)
Tail dropped	Grabbed, frightened, or a cagemate	Paper towel, keep it clean, no handling for 1 to 2 weeks (page 19)
Weight loss between log entries	Several possible causes	Vet, bring the Owner Log (page 30)

Daily, weekly & seasonal routine

Daily

- ☐ Fresh water, and check the humid hide is still damp
- ☐ Feed in the evening on the schedule for the gecko's age, and remove uneaten insects
- ☐ Quick visual check: tail thickness, eyes clear, stool present and normal, no stuck shed
- ☐ Look for the gecko at dusk. A gecko that is not out and moving in its own activity window is the earliest soft signal there is

Weekly

- ☐ Weigh in grams and log it on page 30
- ☐ Verify the warm hide floor with the infrared gun, and read the hygrometer
- ☐ Replace the humid hide moss if it has started to break down, and top up the calcium dish
- ☐ Scrub the water dish rather than just refilling it

Monthly

- ☐ Deep clean: strip and replace substrate, scrub hides and dishes with reptile-safe disinfectant
- ☐ Review the Owner Log for slow trends, not just single readings

Every 3 months

- ☐ Check supplement expiry dates and replace anything opened over a year ago

Seasonal

- ☐ Autumn: expect appetite to drop. Weigh every two weeks through any brumation
- ☐ Winter: check the night low in the room, and add a thermostatted ceramic emitter if it falls below 60°F (16°C)

Yearly

- ☐ Wellness exam and a fecal test
- ☐ Replace the UVB tube, even if it still lights
- ☐ Re-read pages 22 and 23 against the actual setup. Things drift

IF YOU ONLY KEEP ONE HABIT

Keep the weekly weigh-in. Almost everything in Section 03 shows up as a weight change before it shows up as a symptom, and a gecko is small enough that a 5 g slide is invisible by eye and obvious on a scale. Sixty seconds, once a week, on a kitchen scale.

WHAT A GOOD WEEK LOOKS LIKE

Warm hide holding 88 to 92°F (31 to 33°C) every time you check it. Weight flat or climbing. Stool passed on the days it was fed, firm and dark with a soft white urate. A gecko that comes out at dusk and takes food from tongs without hesitating.

Power outages, travel & transport

A leopard gecko's normal night low is around 70°F (21°C). Act once it is heading below 60°F (16°C), and remember a small gecko loses heat faster than a heavy-bodied snake in the same room.

When the power goes out

Do

- Put a battery or wind-up thermometer in the enclosure so you know the real number rather than guessing.
- Drape thick towels or a blanket over the enclosure, leaving a vent gap, and move it to the warmest interior room.
- Set a wrapped hot water bottle or a commercial hand warmer beside the gecko, never touching it, and stop feeding insects until heat is reliably back.

Don't

- Don't use candles, gas stoves, or open flame anywhere near the enclosure.
- Don't put a heat pack in direct contact with the gecko. A cold, sluggish animal cannot move away fast enough and will burn.

OUTAGE KIT

Two or three hand warmers or a hot water bottle, two thick towels, a battery thermometer, a ventilated tub, and both vet numbers written on paper.

Leaving town

TRIP LENGTH	ADULT	JUVENILE
1 to 2 nights	Usually fine alone. Lights on timers, thermostat running, fresh water and a damp humid hide before you go.	A sitter once a day for food, water and a temperature check.
3 to 7 nights	A sitter every 1 to 2 days for water, a stool check and a temperature glance. Feed on the normal schedule.	A sitter daily, with pre-portioned, pre-dusted insects and written instructions.
Over a week	A daily sitter who has done a practice visit with you, or boarding with a reptile-experienced vet or specialist shop.	

Transport and vet trips

- A ventilated plastic tub with a towel to grip beats a wire carrier: it holds heat and stops the gecko watching the world rush past. No loose substrate, no open water dish.
- Skip the meal before travelling, warm the car first, and for any trip over about 30 minutes below 65°F (18°C) pack a wrapped hand warmer beside the tub, not against the animal. Check the tub at a stop every 20 to 30 minutes.
- Never leave the gecko in a parked car, hot or cold, even briefly.

FILL IN AND LEAVE BY THE ENCLOSURE

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Gecko's name _____

Age and normal weight _____

My phone _____

Reptile vet _____

24-hour emergency clinic _____

Backup contact _____

Each visit

- | | |
|--|---|
| ☐ Check the warm hide reads 88 to 92°F (31 to 33°C) and the lights are on their timer | ☐ Offer the pre-portioned insects on the days marked, and remove any left after 15 minutes |
| ☐ Fresh water in the shallow dish | ☐ Remove any stool and note whether it looked normal |
| ☐ Squeeze the humid hide moss, and dampen it if it is dry | ☐ Wash your hands before and after. Don't rinse anything in the kitchen sink. |

Feeding while I'm away

Feed on these days _____

Insects are kept _____

How many insects _____

Dust with (which tub, which days) _____

Please don't

- Take the gecko out of the enclosure, or pick it up by the tail under any circumstances. It will drop the tail.
- Change any dial, timer, or thermostat setting, or move the lights.
- Feed anything not on this sheet.

CALL ME IF

The warm hide reads below 70°F (21°C), or the lights are off when they should be on. There is no stool across two visits. The gecko is breathing with its mouth open, wheezing, or is limp and hard to rouse. Anything is protruding from the vent. The tail has come off. If I don't answer, call the vet number above.

Owner log

A steady weight trend is the single best early warning system you have, and in this species it is doubly true: the difference between brumation and cryptosporidiosis is visible in the log weeks before it is visible in the gecko. Sixteen rows is four months, long enough to see a real trend.

Equipment, supplement & vet log

Equipment log

ITEM	INSTALLED	REPLACE OR CHECK BY	DONE
UVB tube (replace every 12 months)			
Heat mat and thermostat (verify monthly)			
Humid hide moss (replace when it breaks down)			
Thermometer and hygrometer batteries			
Supplements opened (replace yearly)			

Twelve-month planner

Tick the box in the month you did it. Start with whichever month you're in.

TASK	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Deep clean	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Verify warm hide temperature	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Supplement expiry check	☐			☐			☐			☐		
Vet exam	☐											☐

Vet visit log

DATE	WEIGHT (G)	REASON AND FINDINGS	FOLLOW-UP

Enrichment checklist & log

Setup

- ☐ All three hides in place, snug rather than roomy, and the humid one genuinely damp
- ☐ Cork flats, slate or a low branch to rest on and rub against
- ☐ A dig box of damp safe substrate, especially for a female

Monthly review

- ☐ Rotate or move one item, not the whole layout
- ☐ Vary the feeding presentation: tongs, a short supervised hunt, a rolling dispenser
- ☐ Try one new safe feeder insect from page 10

Enrichment log

DATE	ACTIVITY	HOW IT WENT, AND WHETHER TO REPEAT

ENRICHMENT SAFETY

Never leave feeder insects loose in the enclosure overnight, never use wild-caught insects, and never treat a shed cycle as handling time. Nothing on this page substitutes for correct belly heat and three working hides, which is what the research on this species keeps pointing back to.

Glossary

Autotomy	Voluntary tail loss. In a leopard gecko the tail regrows in 4 to 8 weeks, shorter and thicker than the original.
Brumation	A reptile's winter slowdown: reduced activity and appetite triggered by cooler temperatures and shorter days. Not hibernation, and not weight loss.
Crepuscular	Most active at dusk and dawn, which is why evening is the right time to feed and handle.
Cryptosporidiosis	A contagious parasitic infection with no cure, known as stick tail disease for the wasting it causes. Page 17.
Dysecdysis	An abnormal or incomplete shed. Usually a humidity problem, occasionally a vitamin A one.
Dystocia	Egg binding: a female unable to pass eggs she has produced. A same-day veterinary problem. Page 15.
Ecdysis	The normal shed cycle, from the dull pre-shed phase through to the skin coming off.
Ectotherm	An animal that takes its body heat from its environment. It is why the warm hide reading is a health number, not a comfort setting.
Gut-loading	Feeding a feeder insect a calcium-rich diet for 24 to 48 hours before it is offered, so the gecko eats the contents too. Page 11.
Hemipenal bulge	The paired swelling behind the vent that marks a male. Page 14.
Humid hide	A closed box of damp substrate at 70 to 80% humidity inside, in an enclosure that is otherwise 30 to 40%. The single most useful furnishing in this setup.
Impaction	A blockage of the digestive tract, usually from swallowed substrate or oversized prey. Page 18.
Metabolic bone disease	Skeletal weakening from chronic calcium or vitamin D3 deficiency. Preventable, and permanent once advanced. Page 17.
Obligate insectivore	An animal that must eat insects and cannot use plant food at all. That is a leopard gecko.
Oxalic acid	A compound in spinach, beet greens and chard that binds calcium into an unusable form, which is why they are the wrong gut-load. Page 11.
Preanal pores	The V-shaped row of pores in front of the vent, enlarged in males and used for scent marking. Page 14.
Thermal gradient	The warm-to-cool range across the enclosure that lets a gecko choose its own temperature. Without it the animal cannot thermoregulate at all.
Urate	The soft white portion passed alongside stool: a reptile's version of urine. Hard and yellow means dehydrated.
UVI	UV index, the measured strength of UVB at the animal's position. A leopard gecko wants 0.5 to 1.5, and less for a pale morph.

REFERENCE

Sources

Where the numbers in this package come from, so you can check them yourself.

Veterinary and scientific sources

- LafeberVet: Basic Information Sheet, Leopard Gecko. Adult and hatchling weights, sexual maturity, sexing, and common presentations including cystic calculi.
- VCA Animal Hospitals: Leopard Geckos, Care and Feeding.
- PetMD: Leopard Gecko Care Sheet, Stick Tail Disease in Leopard Geckos, and Egg Binding in Reptiles.
- Merck Veterinary Manual: Overview of Reptiles, the husbandry table for selected reptiles, management and husbandry of reptiles, and the parasitic disease page. Thermostat control, the 3 to 6 month quarantine on pages 17 and 25, and paper substrate as the mite-detection surface.
- Gould, A., et al. (2018), *Journal of Herpetological Medicine and Surgery* 28(1-2), 34-39, and Oonincx, D.G.A.B., et al., Wageningen University: cutaneous vitamin D3 synthesis in leopard geckos under UVB.
- Bashaw, M.J., et al. (2016), *Applied Animal Behaviour Science* 184, 150-160, and Rickman, E.L., et al. (2025), University of Lincoln: environmental enrichment and welfare in leopard geckos.
- Egg Removal via Cloacoscopy in Three Dystocic Leopard Geckos (*Eublepharis macularius*), for clutch size and frequency.
- Centers for Disease Control and Prevention: reptiles and amphibians, and the *Salmonella* Muenchen outbreak linked to pet geckos, closed January 2026. Page 8.
- Center for Food Security and Public Health, Iowa State University: Reptile- and Amphibian-Associated Salmonellosis.
- Association of Reptilian and Amphibian Veterinarians (ARAV): find-a-vet directory.

Husbandry references

- ReptiFiles Leopard Gecko Care Sheet, republished by Zen Habitats: temperatures, humidity, UVB output and lighting schedule.
- The Vet Desk: leopard gecko size, weight and growth chart, and cost breakdown.
- Beastly Facts guides, current at publication: the leopard gecko tank setup, temperature, feeding, health, handling, cost and enrichment guides, plus the cross-species reptile heating and thermostats, quarantine, salmonella and hygiene, stool and urates, shedding, gut-loading and emergency plan guides.

Where sources genuinely disagree this package prints the range rather than picking a side: adult length runs from 6.5 to 11 in (17 to 28 cm) across veterinary, species and growth-chart references, hatchling weight from 2 to 8 g, and adult feeding frequency from every other day to every five.

REFERENCE

Version history & about

What changed, and when, so an older printout can be placed.

Version history

VERSION	TEMPLATE	WHAT CHANGED
2.1 · Sep 2026	t3	Cross-checked against the September 2026 cross-species reptile articles. Quarantine raised from 60 to 90 days to the Merck figure of 3 to 6 months on pages 17 and 25, with a separate room rather than a separate enclosure, and paper substrate named as the diagnostic surface it actually is. Page 7 narrows the ceramic-emitter controller advice and adds the daily second thermometer. Page 8 carries the 2026 CDC gecko outbreak and the full hygiene rules. Page 21 names the urate and kidney problem behind a discolored urate, and the soak went to 15 to 20 minutes. Sources and version history split onto their own pages.
2.0 · Sep 2026	t3	Rebuilt and expanded from 22 to 34 pages. Metric units throughout, a thermostat and probe page, feeder and gut-load charts, refusal thresholds, sexing and growth, females and egg binding, split condition pages, poop and hydration, an outage and travel plan, a pet-sitter sheet, longer logs, a glossary and this page. Calcium with D3 corrected to a backup for setups without UVB, not a routine dusting.
1.0 · 2026	pre-t3	First edition, 22 pages, on the earlier layout.

PAGE 7 IS THE PAGE THE SITE ARTICLE AGREED WITH

The September 2026 cross-check compared every reptile package against the new heating and thermostats guide. This one's probe instruction, flat on the warm hide floor on the surface the gecko lies on, taped so it cannot move, matched the article almost word for word and needed no correction. It is the reason page 7 exists and it is the page that prevents burns.

This package is a husbandry reference compiled from the sources on page 34 and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for a reptile-experienced veterinarian, and nothing in it should delay a vet visit. Free companion guides are at beastlyfacts.com.

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